Compatible affine contraction plans for 3-SAT

A stricter certified exponential bound, exact finite certificates, and the remaining logarithmic bottleneck

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Abstract

We review the latest 3-SAT research at ProveIt commit `8e76d515063c`, including the five incoming archives added at `26bd83d4b37d`, and the linked Anthropic formalization. The smallest incoming general-degree cached contraction coefficient is 29.81160894538347, on the scale $s/\sqrt{\log_2 s}$, where s is a normalized tensor-network wire budget. We reduce this coefficient to 29.1489213688721, with a safe eventual coefficient 29.149.

The new construction uses twelve ten-group contraction plans obtained by adding a tenth point to the parallel lines of the affine plane over $\mathbb{F}_3$. Its proof keeps one operation choice consistent across every pairwise comparison. An exact integer elimination and branch-and-bound certificate checks all 96^{12} support choices without enumerating them. A separately written checker reconstructs the supports and recomputes the search bounds. A sharper, already published rectangular multiplication envelope gives the final coefficient; a separate exhaustive certificate improves the next recursion stage. The new plans alone give 29.640645 with the previous backend.

This is an improvement of an explicit asymptotic upper bound, relative to the imported contraction and encoding lemmas. It does not establish a smaller unrestricted 3-SAT exponent order: the strongest supported order remains $n \log\log n/\sqrt{\log n}$, with the existing density-sensitive refinement. No end-to-end solver speedup or new Lean formalization is claimed. We also give a sharp limitation for the preceding nine-plan entropy model and two scoped obstructions encountered while pursuing an order improvement.

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1 Result and comparison target

The requested target is exclusively 3-SAT. Exact counting is used as a subroutine because the reviewed algorithms decide satisfiability by testing whether their count is positive. We keep three quantities separate: the number n of Boolean variables, the number m of clauses, and the wire parameter s of the tensor contractor. A coefficient in the last parameter is not automatically a coefficient per SAT variable.

Theorem 1.1 (New certified contraction bound)**.** *Assume the normalized exact contraction interface described in Section 3: at most s Boolean wires, maximum degree* $\lceil s^{9/10} \rceil$*, the specified coefficient ring and entry access, and the inherited exact packing, deformation, interpolation, and mode identities. There is a deterministic choice of contraction plans with*

$$T(s, \mathcal{L}),\ S(s, \mathcal{L}) \le (s + \mathcal{L})^{O(1)} 2^{(C_* + o(1))s/\sqrt{\log_2 s}},$$

where

$$C_* = \frac{\pi\sqrt{2}}{\ln 2}\sqrt{\frac{20000}{301}\, B_*}, \qquad B_* = \frac{5835847444309}{18750000000000}, \qquad C_* = 29.1489213688721\ldots. \tag{1.1}$$

Here $\mathcal{L}$ *is the description length and all hidden entry-access constants are fixed. The constant* 29.149 *is an eventual upper-bound coefficient for both resources.*

The proof occupies Sections 3–6. The finite inequality behind B_* is proved by a reproducible exhaustive certificate using arbitrary-precision integers. The infinite recursion is handled analytically. The theorem imports the same analytic contractor lemmas used by the preceding packets; it does not require a new unproved mode-summation hypothesis.

1.1 What improved

The incoming pairwise certificate uses nine plans and bounds a quadratic norm by maximizing each plan pair independently. A plan can consequently be charged for different operations in its comparisons with different partners. Our certificate enforces one support choice per plan throughout the whole quadratic expression. This stronger analysis makes a twelve-plan affine construction competitive even though its independent pair bound is worse.

There are two separable improvements:

General-degree cached time bound	Coefficient
Best latest incoming packet: weighted pair certificate	29.8116089454
Consistent choices, same nine plans and supplied weights	29.7483594538
Exact optimum of that nine-plan entropy envelope	29.7460570934
New twelve-plan affine family, previous rectangular backend	29.6406449072
New twelve-plan family, two-anchor published backend	29.5583351755
New family, interior anchor, and improved next stage	**29.1489213689**

The last row lowers the old coefficient by about 0.662688, or 2.223%. These are upper-bound coefficients. Their difference describes the ratio between two certified exponential envelopes, not an observed runtime ratio and not a lower bound on the older implementation. The new algorithm can retain all preceding plans as additional choices; choosing the least certified local cost then includes the new guarantee.

1.2 What remains unresolved

For unrestricted 3-SAT, the supported worst-case rate remains

$$(n+m)^{O(1)}2^{O(n\log\log n/\sqrt{\log n})}. \tag{1.2}$$

The existing density-sensitive refinement is

$$(n+m)^{O(1)}2^{O\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{1.3}$$

At fixed clause density this becomes $(n+m)^{O(1)}2^{O(n/\sqrt{\log n})}$ [1, 4, 5]. Our change improves an explicit contractor constant inside these reductions. It does not remove the $\log\log n$ factor or improve the $\sqrt{\log n}$ denominator. The cached construction uses subexponential space; the previous polynomial-space branch remains available with its previous guarantee.

2 Review of the current repository and formalization

2.1 Frozen snapshot and recent commits

The review freezes ProveIt at

`8e76d515063c4b9fbd8743a846b457e3a01bc039`.

The relevant recent changes, with times reported in Pacific time, are:

Commit	October 26 time	Relevant change
`26bd83d4b37d`	20:51	Five new SAT archives in `docs/incoming`
`7de9d09fecd1`	20:20	Links the public Anthropic Lean development
`a7ae5ab55f96`	20:15	Places nine fourth-round SAT packets
`b9941a7396e4`	20:11	Arrival of those nine packets
`450252e9d86c`	19:36	Places five third-round packets
`6d3adc450615`	19:03	Places eight second-round packets

The project index accounts for twenty-nine refinement packets. The five latest archives are additional and change the current comparison target. We reviewed that index and the recent commit history, read the strongest competing interfaces and proof mechanisms, and examined the five new archives. This is not a claim to have independently rebuilt every earlier packet or checked every proof in the corpus.

Latest incoming archive, abbreviated	Time C_T	Space C_S	Scope
`sat_affine_cauchy`	30.41335	29.46603	General cached
`sat_affine_contraction`	30.25704	25.98907	General cached
`sat_affine_geometry`	30.26987	29.48754	General cached
`sat_affine_plane`	30.69134	29.46603	General cached
`sat_pairwise_certificates`	29.81161	29.81161	General cached

All rows use the normalized $s/\sqrt{\log_2 s}$ scale and the general degree premise. The last archive additionally gives a low-degree polynomial-space coefficient 16.81223, under $\Delta \le s^{1/12}$. The previous general polynomial-space coefficient is 46.18485 [9, 3, 2]. Those smaller low-degree figures have stronger hypotheses and cannot be compared with the general-degree theorem without checking the reduction. The new common space bound does not beat the incoming 25.98907 space tradeoff. All five incoming packets retain the same SAT logarithmic order.

2.2 The linked Lean development is a genuine machine theorem

The user's linked commit is

`anthropics/formal-math@1a6eee293b9e00cb432eef6018a21f523623a9af`.

Its independently stated `MainResults.MainTheorem` quantifies one deterministic Mathlib multi-stack Turing machine with finite stack alphabets, before quantifying k, n, and the formula. It computes the complete satisfying-assignment count within the stated bound

$$2^{C\min\{km/\sqrt{\log_2 m},\ k^2 n \log_2\log_2 n/\sqrt{\log_2 n}\}},$$

subject to the explicit normalization premises: $k \geq 1$, $n \geq 4$, at least two clauses, no repeated clause, and every variable occurring. Deduplication, unused variables, and small inputs need the usual polynomial-time front end. At fixed width three the rate agrees with (1.2) [14, 15].

`Solution.lean` supplies proofs of the public propositions without additional theorem hypotheses. A source scan found the two expected `sorry` occurrences in the separate `Challenge.lean` specification, and no source declarations of additional axioms or unsafe proof shortcuts in the project. The exact linked commit has a successful `verify (eth)` GitHub check. Its README and Comparator configuration describe the standard three Lean axioms and the nanoda check [15, 16].

We inspected these sources and exact-commit CI evidence. We did not independently rebuild the full Lean project or rerun its alternate kernel. Polynomial space is not a conjunct of the exposed Anthropic main theorem, so it must not be inferred from that theorem alone. Its unspecified machine constant also cannot be compared numerically with a normalized wire coefficient. The present contribution is a new finite contraction certificate and plan family, not a revision of that formalization.

3 The exact interface retained from the corpus

The normalized input is a bipartite Boolean tensor network

$$Z = \sum_{x\in\{0,1\}^E} \prod_v T_v(x|_{\delta(v)})$$

with at most s edges, polynomially many vertices, maximum degree at most $\lceil s^{9/10}\rceil$, and description length $\mathcal{L}$. Its entries lie in

$$(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell), \qquad 1 \leq P_0, \ell \leq s.$$

Entry evaluation takes $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ time and polynomial space. The extension ring, immutable creation-time representatives, and exact interpolation conventions are unchanged. In particular, canonical lifting is not assumed to preserve ring operations [1, 6, 3].

At an active call let M be the current wire count, r the fixed group alphabet, u the retained valuation range, and $L = 2^a$ the packet length. At a retained valuation $i < u$ put $D_i = 2^{a-i}$. The imported sharpened estimates give working precision

$$p_t \leq (6 + o(1))MD_i/u,$$

and child wire count $(2/r+o(1))M$. The full-block Cauchy modes have weights $a+1-i+4j-2\,\mathrm{popcount}(j)$, with at most $(2/r^2 + o(1))M/D_i$ families per directed pair at each index. These are inherited lemmas, not new assertions proved by the finite checker [7, 3].

Fix an exact allocation of the weight budget among directed pairs. Let $\mathcal{I}_{gh}$ be the finite index list for a pair, skipping allocations with an empty list. Its contribution is exactly

$$\sum_{\xi_{gh}\in\mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}). \tag{3.1}$$

Each Z_g is a real recursive child contraction, not a unit-cost oracle. For fixed r, allocations and their bookkeeping contribute polynomial factors. Exact knapsack counting and ranking give the cardinalities and indices used below.

Write

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \qquad g_{\{g,h\}} = h_{gh} + h_{hg}, \qquad K(M,u) = \kappa M/\sqrt{u}, \quad \kappa = \frac{\pi\sqrt{2}}{\ln 2}.$$

The common-tilt mode estimate implies, for every fixed nonnegative vector v on unordered pairs,

$$v \cdot g \le \left(\frac{\sqrt{2}\ \|v\|_2}{r} + o(1) \right) K(M,u). \tag{3.2}$$

The $\sqrt{2}$ converts directed to unordered pair coordinates. The error is uniform over the finite family of support vectors and sums used in this paper.

The available asymmetric packing gates, with current side degrees R, C and $Q = r^a$, are

$$QR/M = o(1), \qquad Q^2 C(\log s)^2/M = o(1), \qquad LQ^2/M = o(1). \tag{3.3}$$

The retained side stays under the original degree bound; the fresh side below the root has degree at most $2^{a_{\text{previous}}+1}$.

4 A sharper rectangular operation bound

The cited matrix multiplication result proves $\alpha \ge 0.321334$, $\omega \le 2.371552$, and the interior rectangular bounds $\omega(1, 0.55, 1) \le 2.066134$ and $\omega(1, 0.60, 1) \le 2.092631$ [13, Table 1]. Choose fixed rational values strictly inside the permitted bounds:

$$\alpha_0 = \frac{321333}{10^6}, \qquad \omega_0 = \frac{2371553}{10^6}, \qquad \varepsilon = \frac{1}{10^6}.$$

There are fixed algorithms of rectangular exponent $2 + \varepsilon$ at aspect α_0, exponents $2.066134 + \varepsilon$ and $2.092631 + \varepsilon$ at aspects 0.55 and 0.60, and square exponent ω_0. Every selected exponent has a fixed positive slack above its cited bound. We do not claim these matrix multiplication results as new or as the latest available matrix multiplication bounds.

Let $f(k)$ be the line through $(11/20, 2066135/10^6)$ and $(3/5, 2092632/10^6)$. Tensor products of the two rectangular algorithms realize every rational point of this segment. Put $c = (\omega_0 - 1)/2 = 1371553/2000000$. Intersect $f(k)$ with $1 + c(1 + k)$. Exact arithmetic gives

$$k_* = \frac{59261}{103891}, \qquad b_* = \frac{26972100941}{12986375000} = f(k_*) = 1 + c(1 + k_*).$$

Here $11/20 < k_* < 3/5$, and b_* exceeds the interpolated *published* table bound by exactly ε. Thus the anchor has a genuine fixed-algorithm interpretation.

For sorted logarithmic dimensions $x \ge y \ge z \ge 0$, define

$$\begin{aligned} d &= 1 + \varepsilon/2 = \frac{2000001}{2000000}, \\ w &= \frac{b_* - 2 - \varepsilon}{k_* - \alpha_0} = \frac{7994703637}{25877393297}, \\ v &= 1 + \varepsilon - \alpha_0 w = \frac{2913557133825647}{3234674162125000}, \\ c &= \frac{1371553}{2000000}. \end{aligned} \tag{4.1}$$

In particular $1 > v > c > w > 0$. The operation exponent is bounded by

$$F(x,y,z) = \max\{d(x+y),\ x + vy + wz,\ x + c(y+z)\}. \tag{4.2}$$

Lemma 4.1 (Rectangular bound)**.** *Formula* (4.2) *pays for the matrix operation and its input/output arrays, up to polynomial factors.*

Proof. If $y = 0$, then $z = 0$ and the bound is immediate. Otherwise put $k = z/y$. For $\alpha_0 \le k \le k_*$, tensoring the algorithms at α_0 and k_* gives exponent $2 + \varepsilon + w(k - \alpha_0)$ in the balanced outer dimension y. Blocking the dimension x adds $x - y$, giving $x + vy + wz$. For $k_* \le k \le 1$, the chord from (k_*, b_*) to $(1, \omega_0)$ is $1 + c(1 + k)$; blocking gives $x + c(y + z)$. Both branches are bounded by the maximum in (4.2). Integer roundings of tensor powers affect only fixed factors. If $k < \alpha_0$, padding gives $x+y+\varepsilon y \le d(x+y)$. The largest input or output array has exponent at most $x + y$, covered by the first branch. The bilinear tensor symmetries permit sorting the three logarithmic dimensions. Because $1 > v > w$ and $1 > c$, each branch is the maximum of its coefficient permutations on an unsorted triple. No exchange of minima and maxima is needed. □

We retain the corpus's coordinate/CRT bridge from fixed bilinear algorithms over characteristic zero to the coefficient ring: lift coordinates to bounded integers, use nonexceptional odd primes, reconstruct exact products, and reduce modulo the required power of two. Polynomial rank, precision, and logarithms of the dimensions make this a polynomial overhead. Fixed bilinear recursions may process one child product at a time, with polynomial space overhead relative to the largest input/output array. The bridge is an imported implementation lemma [8]; the finite checker does not execute fast matrix multiplication.

5 Twelve plans from an affine plane

Label nine group vertices by $(x, y) \in \mathbb{F}_3^2$, encoded as $3x + y$, and add vertex 9. Use the four linear forms

$$x, \quad y, \quad x + y, \quad x + 2y.$$

For each form $ax + by$, its three lines are $L_c = \{(x, y) : ax + by = c\}$, $c \in \mathbb{F}_3$. For each of the three choices k, make the ordered partition

$$(L_k \cup \{9\}) \mid L_{c_1} \mid L_{c_2}, \qquad c_1 < c_2, \quad \{c_1, c_2\} = \{0, 1, 2\} \setminus \{k\}. \tag{5.1}$$

Within each line the integer labels are increasing, and 9 is appended to its block. This specifies twelve plans, each with block sizes $4, 3, 3$. Appendix A gives the literal lists.

To build a block with ordered list B, split at $\lfloor |B|/2 \rfloor$, recursively build its two halves, and merge their tables. For a merge with disjoint parts A, B and outside set O, the three index classes are the pairs AB, AO, BO. The final operation contracts the three completed block tables. There are seven internal merges and one final triangle per plan. The singleton tables are computed recursively and charged separately.

5.1 Integer support vectors

The sorted cost (4.2) is exactly the maximum of twelve linear forms on the unsorted triple of dimension logarithms: the three permutations of $(d, d, 0)$, six of $(1, v, w)$, and three of $(1, c, c)$. This follows from the rearrangement inequality. The common integer scale is

$$\mathsf{L} = 51754786594000000.$$

The generating integer triples are

$$\mathsf{L}(d, d, 0), \qquad \mathsf{L}(1, v, w), \qquad \mathsf{L}(1, c, c), \tag{5.2}$$

with exact rational coefficients (4.1). Each is integral; the checker constructs the scale by least common multiples. Assign the coefficient for a class to every unordered group edge in that class, and zero to

edges within a part. Each plan j has an indexed menu $\mathcal{V}_j$ of $8 \cdot 12 = 96$ support vectors in $\mathbb{Z}_{\geq 0}^{45}$; repeated vectors, if any, retain their indices. Define its logarithmic local cost

$$F_j(g) = \mathsf{L}^{-1} \max_{v \in \mathcal{V}_j} v \cdot g.$$

This is the maximum cost of all its matrix operations after the singleton caches are available.

5.2 One consistent choice per plan

Lemma 5.1 (Compatible-support bound)**.** *For q fixed plan menus, suppose*

$$\left\| \sum_{j=1}^{q} v_j \right\|_2^2 \leq B \quad \text{for every } v_j \in \mathcal{V}_j.$$

Selecting the plan of minimum $F_j(g)$ has local time and space exponents at most $(\beta + o(1))K(M,u)$, where

$$\beta^2 = \frac{2B}{\mathsf{L}^2 q^2 r^2}.$$

Proof. For the given g, choose a maximizing support v_j from each menu. Then

$$\min_j F_j(g) \leq \frac{1}{q} \sum_j F_j(g) = \frac{1}{\mathsf{L}q} \left(\sum_j v_j \right) \cdot g \leq \left(\frac{\sqrt{2B}}{\mathsf{L}qr} + o(1) \right) K$$

by (3.2). Every choice belongs to one fixed finite set, so the error bound is uniform. The input/output term in (4.2) pays for every local table; summing a fixed number of table sizes changes only a polynomial factor. □

The preceding pair certificate upper-bounded this norm by independent maxima for each pair of menus. We instead certify its exact global maximum:

Theorem 5.2 (Exact finite affine certificate)**.** *For the twelve menus specified above,*

$$\max_{v_j \in \mathcal{V}_j} \left\| \sum_{j=1}^{12} v_j \right\|_2^2 = 600255570496594046547614542791257241 6. \tag{5.3}$$

Consequently the value $\beta^2 = B_*$ in (1.1) is valid.

5.3 Proof certificate and its completeness

We give the mathematical verification rule explicitly. The supplied programs instantiate it using integers; their searches are finite proof computations, not floating-point estimates of the maximum.

For choices v_i, v_j define the pair potential

$$P_{ij}(v_i, v_j) = \|v_i\|_2^2 + \|v_j\|_2^2 + 2(q-1) v_i \cdot v_j.$$

Expansion gives the exact identity

$$\sum_{i<j} P_{ij}(v_i, v_j) = (q-1) \left\| \sum_i v_i \right\|_2^2. \tag{5.4}$$

Every diagonal term occurs $q-1$ times. For $q=12$, the comparison threshold is eleven times the proposed maximum in (5.3).

Start with all 96 choices in each domain. Given current domains D_j, an upper bound for the pair energy conditional on $v_i=a$ is

$$U_i(a)=\sum_{\substack{j<k\\ i\notin\{j,k\}}}\max_{b\in D_j,c\in D_k}P_{jk}(b,c)+\sum_{j\neq i}\max_{b\in D_j}P_{ij}(a,b), \tag{5.5}$$

where the last expression is interpreted symmetrically. If this is below the threshold, no exceeding tuple can use a. Delete all such choices and repeat. Simultaneous deletions are sound: any exceeding tuple uses only choices surviving the previous round, and its conditional upper bound exceeds the threshold at every coordinate.

After elimination, fix a prefix $v_1,\dots,v_d$. Let E_d be its exact internal pair energy, and for each unfixed j put $u_j(b)=\sum_{i\le d}P_{ij}(v_i,b)$. Then

$$U(\text{prefix})=E_d+\sum_{j>d}\max_{b\in D_j}u_j(b)+\sum_{d<j<k}\max_{b\in D_j,c\in D_k}P_{jk}(b,c) \tag{5.6}$$

is an upper bound on every completion. If it is at most the threshold, the entire branch is certified. Otherwise branch on every remaining choice of the next coordinate. The depth is at most twelve. An unpruned complete tuple is checked exactly. Induction on the number of remaining coordinates proves that successful completion certifies every tuple.

Here the scale and triples are (5.2), pair coordinates are in lexicographic order, operation traversal is postorder, and the twelve triples are sorted lexicographically. These conventions uniquely determine support indices. The witness takes index 88 in every plan. Its directly computed norm equals (5.3), establishing the lower inequality. The upper certificate has:

Certificate quantity	Exact count or value
Support vectors	1152
Pair choices computed	$66\cdot 96^2=608256$
Surviving choices	Two domains of 25; ten domains of 21
Remaining Cartesian choices covered	$25^2\,21^{10}$
Branch-and-bound nodes visited	303513
Maximum norm witness	All twelve support indices equal 88

All remaining branches are certified before depth twelve; the equality witness is evaluated separately. Zero complete leaves visited therefore does not mean the Cartesian product was ignored. The maximum discarded conditional upper bound is

$$660069909523770905947055941890557358 09,$$

strictly below the threshold

$$660281127546253451202375997070382965 76.$$

The first checker constructs incremental conditional sums. The independent checker reconstructs supports differently and recomputes each prefix bound directly. Both use arbitrary-precision integers and explicit exception checks. Substitution into Lemma 5.1 gives B_*. The equality witness has a compact normalized expression:

$$B_*=\frac{47+116c+128c^2}{600}=\frac{5835847444309}{18750000000000}.$$

The expression counts the edges in the sum of the twelve final-operation supports $(c,1,c)$. Its equality with the integer witness is checked exactly; the global upper bound still requires the compatibility certificate.

5.4 Exact plan selection

Let $N_{\{g,h\}} = |\mathcal{I}_{gh}||\mathcal{I}_{hg}|$. For plan j compute the integer

$$Q_j = \max_{v \in \mathcal{V}_j} \prod_e N_e^{v_e}.$$

Minimizing Q_j is exactly equivalent to minimizing $F_j(g)$. The number of factors and their exponents are fixed constants, and the cardinalities have polynomial bit length. This is an exact polynomial-time selector; it requires no oracle for comparing logarithms. The large denominator makes this a theoretical implementation specification rather than a claim of practical efficiency.

6 The complete recursive time and space bound

A local operation bound alone does not prove a faster contractor. Every singleton cache entry is itself a recursive contraction. With $\alpha_r = \sqrt{2(r-1)}/r$, the inherited entry-count estimate and a local plan coefficient β_r give the paid recurrence

$$T_d \le \mathrm{poly}(s)\left[2^{(\alpha_{r_d}+o(1))K_d} T_{d+1} + 2^{(\beta_{r_d}+o(1))K_d}\right], \qquad K_d = \kappa M_d/\sqrt{u_d}. \tag{6.1}$$

We retain the already certified rational schedule from the newest pairwise packet and improve its root and depth-one local operation bounds.

6.1 Integer parameters and packing

Let

$$\lambda = \lceil \log_2 s \rceil, \qquad b = \lceil 6 \log_2 \lambda \rceil, \qquad u_d = \lfloor \eta_d \lambda \rfloor, \qquad a_d = \sum_{j=0}^{d} (u_j + b).$$

Choose

$$\begin{aligned} r_0 &= 10, & \eta_0 &= \frac{301}{20000}, \\ r_1 &= 13, & \eta_1 &= \frac{119}{10000}, \\ r_d &= 7 \quad (d \ge 2), & \eta_d &= \frac{1549}{324000}\left(\frac{4}{9}\right)^{d-2}. \end{aligned} \tag{6.2}$$

The cumulative fractions satisfy

$$\eta_0 + \eta_1 = \frac{539}{20000}, \qquad \sum_{d \ge 0} \eta_d = \frac{8}{225}.$$

Stop at $H = \lceil s/\lambda^{3/4} \rceil$ wires and evaluate directly. Use relative packing slack λ^{-3}. Bounded exceptional sizes are handled by exhaustive contraction after checking the parameter inequalities.

At depth one use the preceding three thirteen-group plans with our anchored backend; in the tail retain the preceding certificate $\beta_7^2 = 632/1225$ [10, 3]. The literal depth-one partitions are

$$\begin{gathered} (0,1,2,3,4) \mid (5,6,7,8) \mid (9,10,11,12), \\ (12,10,1,6,4) \mid (9,0,3,8) \mid (5,2,7,11), \\ (0,8,11,12,1) \mid (7,10,4,3) \mid (5,2,6,9), \end{gathered}$$

and the tail partition is $(0,1,2) \mid (3,4) \mid (5,6)$. Each depth-one plan has ten internal merges and one final triangle, giving $11 \cdot 12 = 132$ supports at the same integer scale L. Exhaustively evaluating all $132^3 = 2299968$ triples gives the exact maximum squared norm

$$B_{13} = 68887464973982598170377980262798003 8.$$

The maximizing operation is the final triangle in each plan, with coefficient triples $(1,c,c),(c,1,c),(c,1,c)$. Its squared norm before integer scaling is $90 + 108c + 198c^2$. Lemma 5.1, now with $q = 3, r = 13$, therefore gives

$$\beta_{13}^2 = \frac{2B_{13}}{\mathsf{L}^2\, 9\, 169} = \frac{20 + 24c + 44c^2}{169} = \frac{57151369949899}{169000000000000}. \tag{6.3}$$

Two independently implemented generators and enumerations reproduce this maximum. Their support indices differ because their traversals differ; the underlying three vectors agree. The tail's older matrix backend remains sufficient. Different fixed matrix algorithms can be used at different recursion levels.

For clarity, here are the strict degree gates that make the schedule valid. The exact integer inequalities

$$10^{1000} < 2^{3322}, \qquad 13^{1000} < 2^{3701}, \qquad 7^{100} < 2^{281}$$

bound the logarithms rationally. The dominant root and retained-side exponents have positive gaps:

Stage	Exponent upper bound	Gap below 1
Root	$9/10 + 2\eta_0(3322/1000)$	39/5000000
Depth one	$9/10 + (539/20000)(3701/1000)$	5161/20000000
Tail	$9/10 + (8/225)(281/100)$	1/11250

The fresh-side and packet-rounding exponents are also bounded away from one, for example by $(8/225)(1 + 2(3701/1000)) < 1$. These fixed gaps absorb the buffers, polylogarithmic losses, and $s/M \le \lambda^{3/4}$.

To avoid assuming the recursion depth in proving the gates, provisionally check them through $3\log_3 \lambda/4 + O(1)$ levels. In this range the smallest taper is at least

$$\lambda^{\delta + o(1)}, \qquad \delta = 1 - \frac{3\ln(9/4)}{4\ln 3} > 0.$$

Nearly lossless packing then makes each tail child smaller than $M/3$ eventually. An active path cannot exceed the provisional bound. Thus the argument closes inductively. Floors are negligible uniformly; padding errors multiply to $1 + o(1)$ over the logarithmic depth. More precisely,

$$M_0 \le s, \qquad M_1 \le (1 + o(1))s/5, \qquad M_d \le (1 + o(1))\frac{2s}{65}\left(\frac{2}{7}\right)^{d-2} \quad (d \ge 2). \tag{6.4}$$

6.2 Precision, entry evaluation, and auxiliary work

At a retained valuation, $D_i \ge 2^{a_{d-1}+b+1}$. The inherited creation-time precision is $O(s\lambda^2 2^{a_{d-1}})$. Relative to MD_i/u_d, it is

$$O(\lambda^{15/4} 2^{-b}) = o(1).$$

Hence inherited precision is absorbed in the sharp local precision estimate at every active depth. Since $a_d \le (8/225)\lambda + O((\log\lambda)^2)$, all precisions have polynomial bit length.

Original entry routines cost $2^{O(s^{9/10})}$ times a polynomial. New packet routines cost $\exp(s^{8/225+o(1)})$. Direct leaf work costs $2^{O(s/\lambda^{3/4})}$. Each is $2^{o(s/\sqrt{\lambda})}$. Immutable descriptions prevent recursive expansion of entry routines. Polynomial overhead at $O(\log\lambda)$ levels contributes only $O(\lambda\log\lambda)$ to the logarithm of the running time. The new support menu and its selector are fixed finite data; their construction does not add a size-dependent exponential term.

6.3 Unrolling every branch

On the scale $s/\sqrt{\lambda}$, set

$$K_0^* = \frac{\kappa}{\sqrt{\eta_0}}, \qquad K_1^* = \frac{\kappa}{5\sqrt{\eta_1}}, \qquad K_2^* = \frac{2\kappa}{65\sqrt{1549/324000}}.$$

In the tail the ratio is $\gamma = (2/7)/\sqrt{4/9} = 3/7$. Moreover $\alpha_7/(1-\gamma) = \sqrt{3}/2 > \beta_7$, so the complete tail branch dominates any tail local termination. Unrolling (6.1) therefore gives the maximum of

$$C_0 = \sqrt{B_*}\, K_0^* = C_* = 29.1489213688721\ldots, \tag{6.5}$$

$$C_1 = \alpha_{10}K_0^* + \beta_{13}K_1^* = \frac{\pi}{\ln 2}\left(\sqrt{\frac{7200}{301}} + \sqrt{\frac{800\beta_{13}^2}{119}}\right) = 29.0008821742914\ldots, \tag{6.6}$$

$$\begin{aligned} C_\infty &= \alpha_{10}K_0^* + \alpha_{13}K_1^* + \frac{\alpha_7 K_2^*}{1-\gamma} \\ &= \frac{\pi}{\ln 2}\left(\sqrt{\frac{7200}{301}} + \sqrt{\frac{19200}{20111}} + \sqrt{\frac{77760}{261781}}\right) = 29.0657485246412\ldots. \end{aligned} \tag{6.7}$$

The exact rational interval checker proves $C_0 > C_1, C_\infty$ and

$$29.1489213688721 < C_0 < 29.1489213688722 < 29.149.$$

It bounds π by alternating arctangent series, $\ln 2$ by its positive series at $1/3$, and square roots by integer squares. No floating-point comparison is used in acceptance.

Local cache sizes are bounded by $\sqrt{B_*}K_0^*, \beta_{13}K_1^*, \beta_7 K_2^*$. The root is largest. Tables on an active ancestor path coexist, but their sizes add; their leading exponent is the maximum, not the sum. The fixed matrix recursion uses polynomial overhead relative to its largest arrays. This proves the common time and space coefficient in Theorem 1.1.

6.4 Consequences for 3-SAT

The direct bounded-density clause network and the sparse polynomial-query reduction both retain their exact counting semantics. For the latter, the original retained incidence degree is $O(n)$, but its compressed Boolean-wire count is $O(n \log\log n)$. Substituting the smaller contractor coefficient therefore improves the same normalized contraction guarantee inside that reduction. It does not turn s into n with constant one. An explicit per-variable numerical coefficient would require all encoding constants and outer costs to be made explicit under the same normalization.

The new cached branch can coexist with the former polynomial-space and lower-space branches. A user who imposes a memory limit should select the applicable resource guarantee; the lowest time coefficient alone does not determine the best implementation.

7 What further constant tuning can and cannot do

7.1 Exact optimum of the preceding nine-plan envelope

This subsection uses the preceding packet's rectangular backend, integer scale 1400000, and its nine plans. It helps distinguish a new plan construction from merely reweighting the old certificate. Let a_i be the chosen support with index

$$(67, 67, 69, 69, 67, 69, 67, 69, 69)_i,$$

using the supplied lexicographic conventions. Put $G_{ij} = a_i \cdot a_j$, $z = G^{-1}\mathbf{1}$, and $p = z/(\mathbf{1}^{\mathsf{T}}z)$. The accompanying rational checker proves that G is invertible and all $p_i > 0$. Define

$$H = \frac{1}{1400000^2\,\mathbf{1}^{\mathsf{T}}G^{-1}\mathbf{1}}.$$

With these weights, conditional pair elimination leaves 1536 support tuples. Their exact maximum weighted squared norm is H.

Proposition 7.1 (Restricted nine-plan optimum)**.** *For the old nine menus with their old backend, the worst-case value of* $\min_i F_i(g)$ *over nonnegative unit Euclidean vectors* g *is exactly* $\sqrt{H}$*. At the retained root schedule its coefficient is strictly between* 29.746057 *and* 29.746058*.*

Proof. The weighted version of Lemma 5.1, verified globally with the rational p, gives the upper bound. For the lower bound put

$$b = \sum_i p_i a_i/1400000.$$

The equation $Gp = \mathbf{1}/(\mathbf{1}^{\mathsf{T}}z)$ implies

$$\|b\|_2^2 = H, \qquad (a_i/1400000)\cdot b = H \quad \text{for every } i.$$

All coordinates of b are nonnegative. At $g = b/\sqrt{H}$, each plan has a support of value $\sqrt{H}$, so $\min_i F_i(g) \geq \sqrt{H}$. Together with the upper bound this proves equality. □

This lower bound belongs to the Euclidean support envelope. It is not a SAT lower bound and need not be attained by actual generated mode lists. It nevertheless proves that further manipulation of mixing weights, even depending on the support tuple, cannot improve that unchanged nine-plan envelope below the displayed value. The new affine plans escape the premise by changing the menu.

7.2 A uniform-entropy limit for this twelve-plan architecture

For any ten-group plan ending in blocks of sizes $4, 3, 3$, uniform unordered edge entropy gives final logarithmic dimensions in the ratio $12 : 12 : 9$. With the anchored backend (4.2), the prescribed cost envelope of the last operation is at least the linear branch $12 + 21c$ times the entropy of one edge. Consequently its Euclidean-envelope coefficient obeys

$$\beta \geq \frac{\sqrt{2}}{10\sqrt{45}}(12 + 21c). \tag{7.1}$$

This holds for every menu of such plans, because uniform entropy makes all final partitions identical in dimension. At $\eta_0 = 301/20000$, the resulting root coefficient is at least

$$\frac{\kappa}{\sqrt{301/20000}}\,\frac{\sqrt{2}(12 + 21c)}{10\sqrt{45}} = 29.0807310194429\ldots.$$

Thus this backend, root schedule, and final block architecture leave only about 0.0682 absolute coefficient headroom below C_*. Changing just these menus cannot reach the unchanged tail branch $29.0657485\ldots$. This is a useful stopping criterion for that particular search, not a limitation on other contraction architectures or improved matrix backends.

8 Attempts to change the logarithmic order

The main square-root scale comes from the mode entropy $K(M,u) = \Theta(M/\sqrt{u})$ together with admissible $u = O(\log s)$. A fixed finite improvement of a positive local coefficient keeps that scale. The extra $\log\log n$ in the dense SAT reduction is the information needed by the current selected-incidence encoding. It is not a polynomial arithmetic factor that disappears by using a faster multiplication routine.

Several apparent order improvements were already explored in earlier packets. Retuning the Hermite taper, imposing the ordinary-degree cap alone, applying the same contractor directly to encoded mode indices, and optimizing ordinary sparsification against the current density cost do not supply the missing recursive interface [11, 12]. We did not repackage those conditional routes as new theorems.

Two additional elementary obstructions emerged from the present attempts. Their hypotheses are deliberately narrow.

8.1 Changing the prime requires a different packet geometry

Lemma 8.1 (Two good cyclic shifts)**.** *Let p be prime and $R, C : \mathbb{Z}/p\mathbb{Z} \to A$ be row and column group labels. A shift d is good if $R(t) = C(t+d)$ for every t. If two distinct shifts are good, both labelings are the same constant.*

Proof. For good $d \neq e$, $C(t+d) = R(t) = C(t+e)$. Thus $C(s) = C(s+e-d)$ for every s. The nonzero difference $e-d$ generates the cyclic group of prime order, so iterating this equality makes C constant everywhere. Then $R(t) = C(t+d)$ gives the same constant. □

For $p = 2$ there is one nonzero shift, so a nonconstant swapped labeling can localize every nonzero residue. For an odd prime, two such residues already force constancy. A nonconstant prime-cycle construction can retain at most one of the $p-1$ nonzero residues as good. Thus the observation that a larger prime reduces the number of precision digits does not by itself generalize the binary packet proof.

The precise finite-group extension identifies an escape: for a finite abelian group G and a nonempty set D of good shifts, C need only be constant on cosets of $\langle d - e : d, e \in D\rangle$. Conversely that condition, with $R(t) = C(t+d_0)$ for a fixed $d_0 \in D$, makes every shift in D good. This follows by invariance under the generators in one direction and under $d - d_0$ in the other. Affine vector spaces and other packet geometries are therefore not ruled out by Lemma 8.1.

8.2 Coherent original-variable substitutions do not automatically split

Lemma 8.2 (Incidence coherence)**.** *Give variables and factors group labels which agree at every factor–variable incidence. Then every connected component of the bipartite incidence graph has a single group label.*

Proof. The labels at the endpoints of every edge are equal. Equality propagates along a path, and any two vertices of a connected component are joined by a path. □

Corollary 8.3 (Shared linear-form version)**.** *Let $E = \bigoplus_{j=1}^{r} E_j$. Replace every occurrence of original variable v by the same nonzero vector or linear form $\ell_v \in E$. Suppose factor f is assigned to group $\sigma(f)$, and locality is enforced by $\ell_v \in E_{\sigma(f)}$ for every variable in its scope. Then all factors in a connected incidence component lie in one group. If its k variable images are also linearly independent, that group has dimension at least k.*

Proof. If a variable met factors in two different groups, its image would belong to their intersection, which is zero, contradicting nonzeroness. The incidence lemma then gives one group per connected component. Linear independence of k vectors in that group implies the dimension bound. □

This rejects only a direct, fully coherent substitution that enforces locality by structural containment. It permits rank-deficient or vanishing images, factor-specific cancellations, measurements, correction modes, and occurrence-dependent substitutions. The source's measurement machinery falls outside the restrictive hypothesis. The corollary is not a barrier to all degree-preserving encodings.

8.3 Decision alone does not justify tiny parity-check budgets

Reducing the precision of the final answer is not enough to remove the linear term in the number of parity tests. The sparse-query degree bound already contains a term proportional to the number of tests, even when the requested output precision is small. A useful scoped example is a formula with disjoint clauses: every Boolean clause-violation vector is realizable. If a linear parity map accepts exactly the zero violation vector, its kernel must be trivial, so its rank equals the number of clauses. This does not make disjoint clauses difficult to solve directly. It only invalidates a general argument that decision automatically permits sublinearly many exact parity filters.

We have not proved a new smaller-order deterministic 3-SAT algorithm. No experiments in this archive are evidence for such a claim.

9 Verification, reproducibility, and proof status

The archive separates imported analytic theorems, new finite inequalities, and finite semantic examples. This separation matters: a correct integer certificate for a support norm does not by itself implement the tensor contractor or prove its deformation identities.

Obligation	Evidence and scope
Root support maximum	Two distinct support generators and two distinct branch-bound implementations reproduce the exact norm, elimination domains, and 303513 search visits.
Thirteen-group maximum	Two independent implementations enumerate all 2299968 triples and find the same unique maximizing vectors.
Backend arithmetic	Rational interpolation, positive endpoint slack, permutation supports, and the integer scale are checked exactly. The existence of fast bilinear algorithms is imported from the cited paper.
Recursive coefficient	Exact integer logarithm comparisons, rational packing margins, and rational intervals for π, $\ln 2$, and square roots certify every branch and the bound $C_* < 29.149$.
Contraction semantics	Eight scalar factor-graph fixtures over $\mathbb{Z}/8\mathbb{Z}$, with up to 4096 assignments, compare all twelve plans to direct enumeration: 96 equalities.
Selector semantics	The finite prototype checks exact integer scores for dyadic dimensions, bypasses selection on empty mode lists, and rejects negative dimensions.
Prior-menu limitation	A rational Gram-matrix certificate, positive weights, an exhaustive residual support check, and a matching direction prove Proposition 7.1.
Order obstructions	The proofs in Section 8 are elementary. Small exhaustive checks illustrate them and include composite-group examples to test the stated scope.

The final acceptance conditions use explicit exceptions rather than Python assertions, and the principal independent checks also pass with `python3 -O`. Numerical displays are not acceptance tests. The scripts use only the Python standard library.

9.1 How to reproduce the supplied computations

From the extracted archive root run:

```
python3 verify_all.py
python3 -O verify_all.py
```

The second command is an optional repeat demonstrating that optimized Python does not remove acceptance conditions. A shorter review is available as `python3 verify_all.py -quick`; it skips the separate root replay and is not a replacement for the complete audit. The wrapper isolates generated files from the shipped reference results. The root certificate can also be checked directly with

```
python3 code/independent_affine_audit.py \
  code/anchored_backend_certificate.json
```

The local README documents outputs and the other individual scripts. The supplied PDF can be rebuilt, given a LaTeX installation, with `python3 build_report.py`. The archive includes all referenced `.tex` sections; no remote inputs are needed for compilation.

9.2 What is not certified by this archive

There is no new Lean formalization of the plan or norm certificate. The programs do not implement the asymptotically fast matrix algorithms, the complete exact interpolation contractor, or an end-to-end 3-SAT solver. Their finite scalar timings would not predict the practicality of those asymptotic algorithms. No threshold for the eventual constant is asserted: the packing margins are small, the fixed matrix algorithms are theoretical, and the hidden constants remain large.

The mathematical theorem is relative to the explicitly named contraction, mode-counting, reduction, and ring-implementation lemmas of the reviewed corpus. The full cited formalization was inspected at the statement and source level, with its exact-commit CI status, but not rebuilt locally. These are material boundaries of the present verification.

10 Next targets that could change the order

The most useful next search should change an interface that produces the logarithmic loss. More finite menus within the same envelope can improve the coefficient, but Section 7 already quantifies the limited remaining room for the present root architecture.

Original-variable measurements with paid shared information. A promising target is an exact decomposition which retains original variable directions while exposing shared measurements of lower total description entropy than the current selected-incidence indices. Its theorem would need to specify the reconstruction identity, coefficient access algorithm, precision, child degree, child wire count, and the cost of enumerating or contracting the shared measurements. The incidence coherence lemma explains why a fully shared linear substitution without such measurements is insufficient. This is a research interface to construct, not an algorithm established here.

A different graded exact crossing representation. The Anthropic paper already identifies the growth of the graded crossing space as a possible source of a different denominator power. Ordinary partition growth gives the square-root scale. A replacement with smaller graded growth would still have to satisfy the complete locality and duality identities and efficient coefficient access. Truncating or regrading the same oscillator representation does not supply these properties; earlier packets audit those restricted attempts [17, 18]. This is an existing open route, not a new improvement claimed by this memorandum.

A new packet geometry in another characteristic. The finite-abelian-group version of Lemma 8.1 identifies precisely where the prime-cycle obstruction can be escaped. A viable construction would still have to preserve enough interpolation nodes, pay the jet-recovery loss, obtain an integral separated kernel, and close the recursion. A lower initial precision alone does not meet those requirements.

The proved advance in this archive is the combination of a new plan family, compatible support choices, a sharper rectangular envelope, and the paid recursive certificate. The requested smaller logarithmic order for unrestricted 3-SAT remains open in this work.

A Literal root plans

The following lists are ordered. Midpoint splitting and postorder operation traversal use exactly these orders. Plan numbers are zero based. The first three plans correspond to the form x, the next three to y, then $x+y$, then $x+2y$. Within each set of three, the chosen line value is $0, 1, 2$.

Plan	Four-vertex block	First triple	Second triple
0	$(0,1,2,9)$	$(3,4,5)$	$(6,7,8)$
1	$(3,4,5,9)$	$(0,1,2)$	$(6,7,8)$
2	$(6,7,8,9)$	$(0,1,2)$	$(3,4,5)$
3	$(0,3,6,9)$	$(1,4,7)$	$(2,5,8)$
4	$(1,4,7,9)$	$(0,3,6)$	$(2,5,8)$
5	$(2,5,8,9)$	$(0,3,6)$	$(1,4,7)$
6	$(0,5,7,9)$	$(1,3,8)$	$(2,4,6)$
7	$(1,3,8,9)$	$(0,5,7)$	$(2,4,6)$
8	$(2,4,6,9)$	$(0,5,7)$	$(1,3,8)$
9	$(0,4,8,9)$	$(2,3,7)$	$(1,5,6)$
10	$(2,3,7,9)$	$(0,4,8)$	$(1,5,6)$
11	$(1,5,6,9)$	$(0,4,8)$	$(2,3,7)$

B Exact constants for independent transcription

The two principal finite certificates reduce to

$$B_* = \frac{47+116c+128c^2}{600}, \qquad \beta_{13}^2 = \frac{20+24c+44c^2}{169}, \qquad c = \frac{1371553}{2000000}.$$

The compact root coefficient is

$$\boxed{C_* = \frac{\pi}{\ln 2}\sqrt{\frac{23343389777236}{564375000000}}}.$$

The coefficient comparison uses the same rational retained fractions 301/20000, 119/10000, and the geometric tail in (6.2). Changing these fractions without rechecking packing, precision, and recursive branch costs is not justified by either finite certificate alone.

References

[1] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026 preprint, especially quantitative contraction and quantitative reduction appendices. Vendored in ProveIt at `8e76d515063c`; upstream pin `7ef5c0cc5ca4`. Pinned preprint directory.

[2] Vladimir Reshetnikov, ProveIt, `docs/incoming`, five archives added at commit `26bd83d4b37d`, October 26, 2026 Pacific time. Pinned incoming directory. Archive hashes and comparison premises are in the accompanying provenance.

[3] *SAT pairwise certificates*, October 26, 2026 research packet, `sat_pairwise_certificates_2026-10-26.zip`. Pinned archive. Baseline nine-plan certificate and strict three-stage schedule.

[4] *Density-aware SAT improvement*, ProveIt research packet, `Research/sat_density_improvement`. Pinned theorem source.

[5] *ETH density refinement*, ProveIt research packet, `Research/eth_density_refinement`. Pinned report source.

[6] *Asymmetric packet balancing*, ProveIt research packet, `Research/eth_asymmetric_packet_balancing`. Pinned report source.

[7] *Joint Cauchy contraction*, ProveIt research packet, `Research/eth_joint_cauchy`. Pinned report source.

[8] *Rectangular contraction*, ProveIt research packet, `Research/subsat_rectangular`, especially the rectangular arithmetic and coefficient-ring implementation discussion. Pinned implementation source.

[9] *Adaptive SAT bounds*, ProveIt research packet, `Research/sat_adaptive_bounds`, nested-flag polynomial-space certificate. Pinned certificate source.

[10] *SAT adaptive tree certificate*, ProveIt research packet, `Research/sat_adaptive_tree_certificate`. Pinned report source.

[11] *Method barriers*, ProveIt 3-SAT asymptotic analysis packet. Pinned scoped-barrier analysis.

[12] *3-SAT precision refinement*, ProveIt research packet. Pinned report source.

[13] Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou, *New Bounds for Matrix Multiplication: from Alpha to Omega*, SODA 2024; arXiv:2307.07970v2, especially Table 1 and the tensor-rank interpretation of rectangular exponents. Versioned primary source.

[14] Anthropic, `formal-math`, `eth/MainResults.lean`, commit `1a6eee293b9e` (full hash in Section 2). Exact independent main theorem statement.

[15] Anthropic, same commit, `eth/Solution.lean` and project README. Solution file; Verification documentation.

[16] GitHub Actions, `verify (eth)`, exact-commit successful run. Observed check-run evidence. This memorandum does not claim an independent local Lean rebuild.

[17] Anthropic, accompanying ETH paper at the same pinned commit, particularly the exact crossing-space construction and Appendix H. Pinned paper.

[18] *3-SAT round-two refinement*, alternate representation audit, `research_alternate`. Pinned research directory.